Chapter 1

Introduction to IT Project Management

What is a Project?

- A **project** is a temporary effort undertaken to create a unique product, service, or result.
- An **IT Project** is a project that uses **information technology** as its primary means to create its unique product, service, or result.
  - ✓ *The core work of the project involves the development, implementation, or enhancement of technology systems.*

Project Attributes

- A project is **Temporary**
  - **Every project has a defined beginning and a defined end.**
  - The end is reached when the project's objectives are **achieved**, or when it's **terminated**.

- **A project has a unique purpose.** Every project is undertaken to achieve a specific goal or to solve a problem that is distinct from other activities.

- **Has Unique outcome**
  - **project's product, service, or result is different in some distinguishing way from all other similar products.**
  - While it may be similar to past projects (e.g., building a website), it will have unique elements like stakeholders, technologies, or constraints.
- **Progressive elaboration:** the details are refined and made more specific as the project progresses and more information becomes known.

- ☑ **Requires Resources:** This includes people, hardware, software, and budget.
- ☑ **A project involves uncertainty.** Because every project is unique, it carries an element of risk and uncertainty that must be managed.
- ☑ **A project should have a primary customer or sponsor.** This is the person or group that provides the direction, funding, and ultimate acceptance of the project deliverables.

Key Characteristics of IT Projects

- **The deliverable is a technology system or capability.**
- They often have **high levels of uncertainty and change** due to evolving technology and requirements.
- They require a blend of **technical and business knowledge.**
  - They are particularly prone to risks like **scope creep, technological obsolescence, and integration challenges.**

Examples of IT Projects

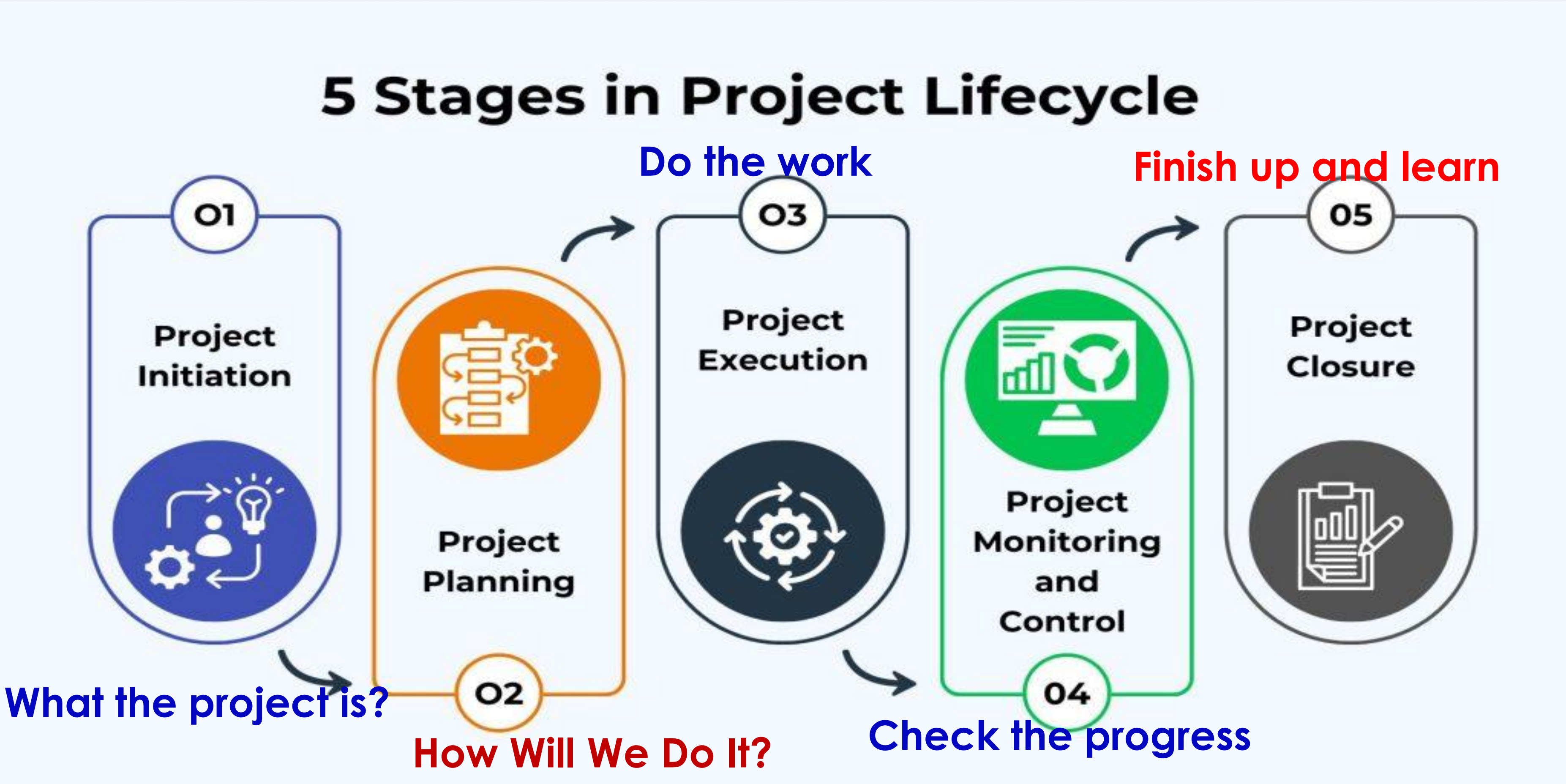
- **Software Development:** e.g. Creating a new mobile application
- **System Implementation:** e.g. Installing and configuring an enterprise resource planning (ERP) system like SAP or Oracle for a company.
- **Infrastructure Upgrade:** e.g. Migrating a company's data and applications from an on-premise server room to a cloud platform like AWS or Azure.
- **Website Redesigning:** e.g. Redesigning and rebuilding a university's public website.
- **Security Enhancement:** e.g. Implementing a new organization-wide cybersecurity and firewall system.
  - **All of these are temporary and aim to create a unique result.**

Project vs. Operations

Feature	Project	Operations
Definition	A temporary endeavor undertaken to create a unique product, service, or result.	The ongoing, repetitive activities that sustain the business and produce the same result repeatedly.
Timeline	Temporary. Has a defined start and end date.	Ongoing / Permanent. Has no planned end date; it is the core business.
Goal	To achieve a specific objective and then terminate.	To sustain the business by producing goods or services efficiently.
Nature of Work	Unique. The work is new and undertaken to deliver something that has not existed before.	Repetitive / Routine. The work follows standardized processes and is performed again and again.
Output	A unique deliverable (e.g., a new software app, a new building, a research finding).	Standardized goods or services (e.g., processing payroll, manufacturing cars, handling customer support calls).
Focus	Effectiveness: "Are we doing the right things?" (Achieving the project goal).	Efficiency: "Are we doing things right?" (Optimizing the process).
Management Approach	Project Management: Manages change, uncertainty, and novelty. Emphasizes planning for a finite end.	Operations Management: Manages stability, consistency, and continuous flow. Emphasizes process optimization.

The Project Life Cycle

A Project Life Cycle is a series of phases that a project passes through from its initiation to its closure.



Initiation

- The primary goal of this phase it to get official approval to start the project-a **signed Project Charter.**
  - What the project is?
  - Why it's important?
  - Who is in charge (the Project Manager).

Planning

"If you fail to plan, you are planning to fail."

- **This is the most intensive phase from a management perspective,** as it involves creating a set of plans that will guide the project through **execution and closure**. This includes the scope, schedule, budget, quality, communications, and risk management plans.
- **Output-Project Management Plan,** which is a comprehensive, integrated document that consolidates all the individual management plans and baselines. It becomes the single source of truth for the project.

Execution

- This is where the conceptual plan is transformed into a **tangible** product, service, or result.
- The longest phase and where the majority of the project budget is spent.
- This Involves coordinating people and resources, managing stakeholder expectations, and conducting procurements.
- The main output of this phase is the project **deliverables themselves.**

Monitoring & Controlling

- This happens at the same time as Execution
- Its purpose is to track, review, and regulate the progress and performance of the project, and to identify any areas where changes to the plan are required.
- **Key activities**
  - Track progress against the schedule and budget.
  - Handle any changes that come up
  - Fix problems before they get worse.

Closure

- **This is the formal conclusion of the project activities.** It is often skipped, but doing so is a major mistake that prevents **organizational learning**.
- **The most critical activity is conducting a Lessons Learned session** to document what went well, what went wrong, and what should be done differently next time.
- **Main activities**
  - **obtaining formal customer or sponsor acceptance of the final product,**
  - **closing all procurement contracts,**
  - **releasing project resources (including the team), and**
  - **archiving all project documents.**

The project project Triple Constraints

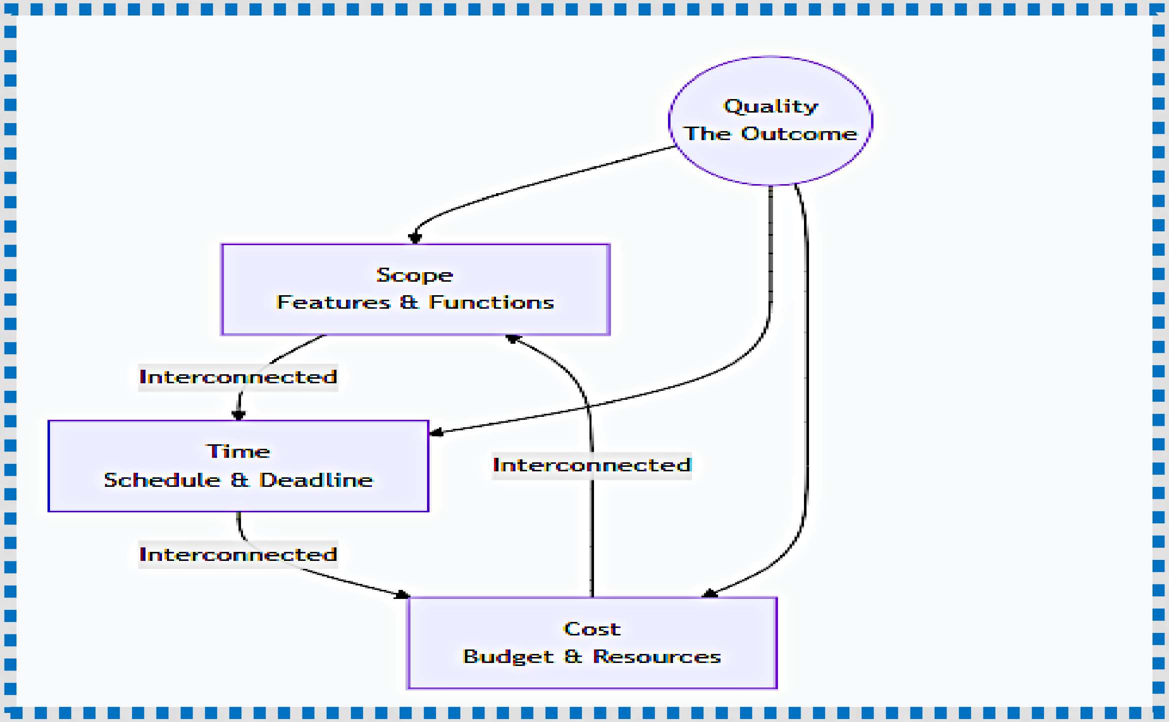
Scope: What work will be done? The list of requirements, user stories, and the specific technical specifications of the software or system

Time: How long will it take? (schedule, deadlines, and the overall timeline for completing the project work, from start to finish.)

Cost: The budget allocated for the project, covering all resources (people, hardware, software, licenses).

Quality: The balance of these three determines the final **Quality.** These three factors are interconnected

The Project Triple Constraint Model



The model states that the **Scope, Time, and Cost** of a project are inseparably linked. We cannot change one without affecting at least one of the others. The quality of the final deliverable is often represented at the center, being a function of the balance between these three constraints.

Scenario

- **Imagine a project to build an e-commerce website with 10 core features in 6 months for $100,000.** This establishes a baseline for scope, time, and cost.
- **Now, the marketing department insists on adding two major new features before launch.** This is a **scope increase**.
- **If the Scope increase**
  - The **Time** and/or **Cost** must **increase** to maintain the same **Quality**.
- **If the time decrease**
  - The **Scope** must decrease and/or increase **Cost** (pay for overtime) to maintain Quality.
- **If the cost decrease**
  - The **Scope** must decrease and/or increase **Time** to maintain Quality.

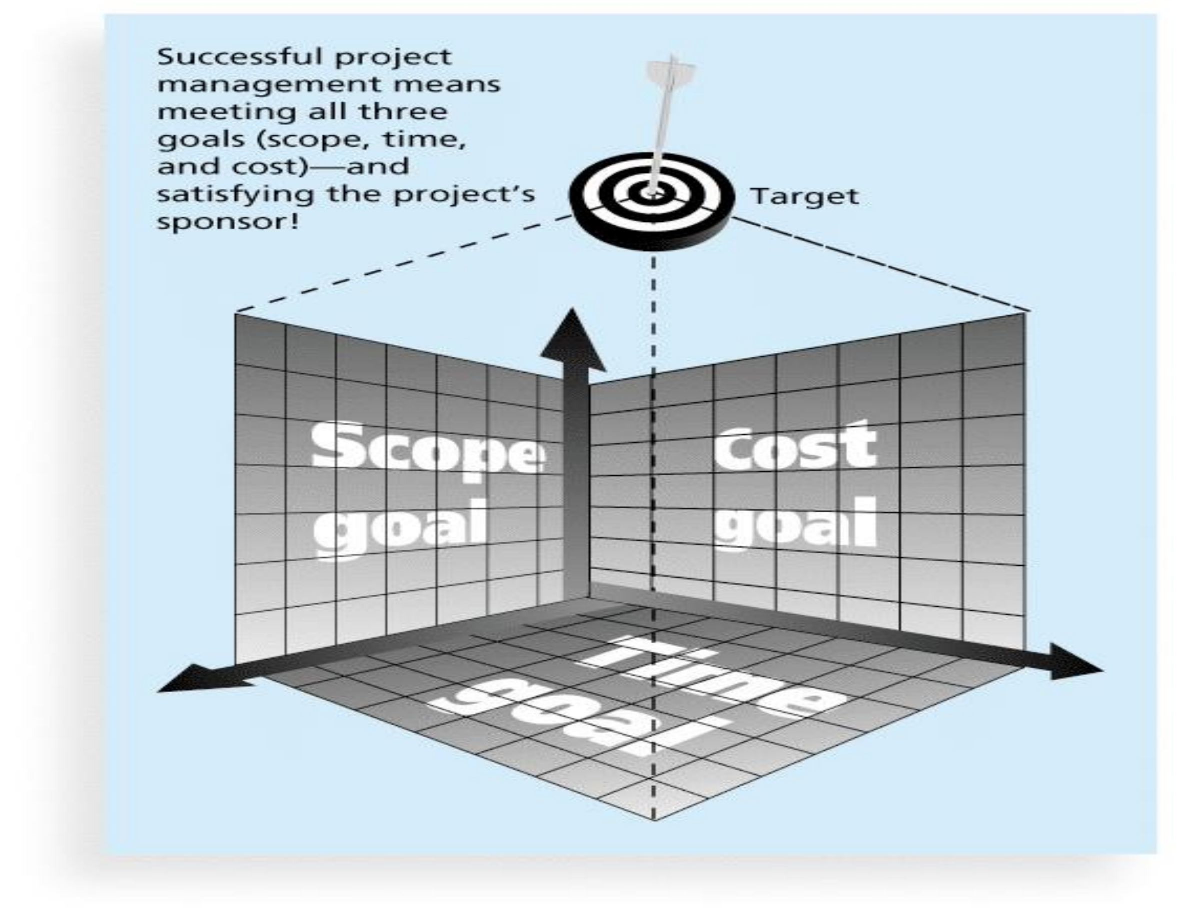
The Project Manager's goal is to balance these constraints to deliver quality and meet stakeholder expectations.

Common Causes for IT Projects Fail

- **Unclear or constantly changing requirements** lead to rework, confusion, and a final product that doesn't meet the users' needs. This is often called "scope creep."
- **Lack of executive support-**the project lacks a powerful champion to resolve conflicts, secure resources, and maintain organizational focus.
- **Inadequate user involvement** leads to a product that is technically good but doesn't work in the real world for the people using it.
- **Poor planning and unrealistic estimation** set the project up for failure from the start, creating impossible deadlines and budgets that the team can never meet.
- **Unmanaged risks** become actual issues that disorganize the project.

What is a Successful Project?

A successful project meets the triple constraint *and* stakeholder expectations.



The Evolution of Project Success

The Traditional View: The Iron Triangle

- Success was narrowly defined by delivering a project:
- **On Time:** Adhering to the original schedule.
- **On Budget:** Staying within the allocated cost.
- **On Scope:** Delivering all the features and functions initially specified.

The Problem: *A project can be on-time, on-budget, and on-scope, but still be a complete failure.*

- For example, if the final product is never used by anyone, delivers no business value, or disrupts the company's operations, was it truly successful?

The Modern View of Success

The modern view adds 2 critical new dimensions to the traditional three

- **Delivering Business Value & Meeting Objectives:** Did the project achieve the strategic business goals outlined in its original business case?
  - Did it increase revenue? By how much?
  - Did it reduce costs? By how much?
  - Did it improve customer satisfaction?
    - **Example:** The success of a new CRM system isn't just that it launched; it's whether sales productivity increased by the projected X%.
- **Stakeholder Satisfaction & Adoption (The "Who")**
  - Are the end-users and sponsors happy with the final product? Is it actually being used?

  Example: A new software tool is delivered on time and budget, but if the **users hate it and refuse to use**, the project has failed.

PROJECT STAKEHOLDERS

- **Stakeholders** are the people involved in or affected by project activities
- Stakeholders include

- the project sponsor
- the project manager
- the project team
- support staff
- customers
- users
- suppliers
- opponents to the project

What is Project Management?

- **The application of knowledge, skills, tools, and techniques to project activities to meet project requirements.**
- it's a multifaceted discipline that integrates leadership, strategy, communication, and technical skills to guide a team through complexity and uncertainty.
- **The ultimate aim is to meet the project requirements-**delivering the agreed-upon scope, within the agreed-upon time and cost, to the satisfaction of the stakeholders.

Project and Program Managers

- **The Project Manager** is the person assigned by the performing organization to lead the team and be responsible for achieving the project objectives.
- **They are a leader, communicator, facilitator, and problem-solver.** They are accountable for the project's success or failure.
- **Program:** A group of **related projects** managed together for greater benefit
- **Program managers** oversee programs; often act as bosses for project managers
- **Portfolio:** A collection of **projects and programs** grouped to achieve strategic business goals such as Company's Entire IT Investment Portfolio.

Project Management Compared to Project Portfolio Management

CAREERS FOR IT PROJECT MANAGERS

Project management is consistently ranked as one of the "hottest" and most in-demand skills in the IT industry, *often appearing at #2 just behind programming/development.*

Job titles

- ***IT Project Manager,***
- ***Program Manager,***
- ***Scrum Master, Portfolio Manager***
- *Chief Information Officer (CIO).*

Skill	Percentage of Respondents
Programming and application development	48
Project management	35
Help desk/technical support	30
Security/compliance governance	28
Web development	28
Database administration	26
Business intelligence/analytics	24
Mobile application and device management	24
Networking	22
Big data	20

Talent Triangle

The Project Management Institute (PMI) defines three key skill sets for the modern project manager.

People & Leadership Skills (Soft Skills)

- Leadership, Team Building, Motivation
- Communication (Verbal & Listening)
- Conflict Resolution & Negotiation
- Integrity and Ethical Behavior

Strategic & Business Skills

- Critical Thinking & Problem Solving
- Understanding Business Strategy
- Balances Priorities

Technical & Analytical Skills

- Project Management Knowledge
- Risk Management & Planning
- Familiarity with PM Tools

Skills for Different Project Types

- **Large Projects:** Leadership, **prior experience**, planning, people skills.
- **High-Uncertainty Projects: Risk management**, expectation management, flexibility.
- **Very Novel Projects: Vision**, self-confidence, listening skills.

Project Management Knowledge Areas

- **Knowledge areas** describe the key competencies that project managers must develop.
- Project managers must have knowledge and skills in all 10 knowledge areas (project integration, scope, time, cost, quality, human resource, communications, risk, procurement, and stakeholder management)

Project Management Framework